universe. This probably would have been a hindrance in the development of life, since it is believed that life as we know it must start in a reducing (low oxygen containing) atmosphere.

Our galaxy has stars much larger and much smaller than our sun, and of widely different ages. Thus the processes we have described are occurring at present in the Milky Way and in other galaxies as well. Moreover, the stars emit considerable amounts of matter into space. As a result, interstellar gas, out of which new stars are formed, contains atoms heavier than helium, although hydrogen is the most abundant element.

12.5. NEUTRON CAPTURE PROCESSES: FROM IRON TO URANIUM

Figure 4.3 shows that the maximum nucleon binding energy occurs at $A \approx 60$, i.e. around iron, which we may consider to be the most thermodynamically stable element in the universe. At lower values of A, fusion of lighter elements releases energy while the exothermic reactions to form heavier elements ($A > 60$) involve neutron capture.

12.5.1. Slow neutron capture

Through hydrogen and helium burning, neutrons are formed. The most important reaction is believed to be

$$^{22}\mathrm{Ne} + {}^{4}\mathrm{He} \rightarrow {}^{25}\mathrm{Mg} + \mathrm{n} \tag{12.33}$$

As the heavier elements form in the star, the neutron production increases considerably since such reactions become more prevalent as heavier elements are involved in the reactions. In reactions like (12.31) and (12.32) energetic γ-rays are emitted, which decompose D into H and n (the binding energy is 2.22 MeV). The mode of production of the elements changes from that of helium capture to that of neutron capture, so that the elements from iron to bismuth can be formed by a slow process of neutron capture (n,γ-reactions), interrupted by β-decay whenever it is faster than the next capture step.

Such a process is known as the *slow*, or *s-process*. While the reaction probability for the capture of neutrons increases with the atomic number of the element, the relative amount of the elements in the star will decrease with increasing atomic number, because of the successive higher order of reaction (§17.3). The result is the observed flattening of the abundance curve for $A > 100$, see Figures 12.5a and 12.10b.

The formation of ^{104}Pd from ^{100}Ru can serve as an example of the steps in the s-process of element formation

$$\begin{array}{l} {}^{100}_{44}\mathrm{Ru}\,(\mathrm{n},\gamma)\ {}^{101}_{44}\mathrm{Ru}\,(\mathrm{n},\gamma)\ {}^{102}_{44}\mathrm{Ru}\,(\mathrm{n},\gamma)\ {}^{103}_{44}\mathrm{Ru} \\ \qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\beta\downarrow 39\ \mathrm{d} \\ \qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad {}^{103}_{45}\mathrm{Rh}\,(\mathrm{n},\gamma)\ {}^{104}_{45}\mathrm{Rh} \\ \qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad (\mathrm{stable}) \\ \qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\beta\downarrow 42\ \mathrm{s} \\ \qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad {}^{104}_{46}\mathrm{Pd}\ (\mathrm{stable}) \end{array} \tag{12.34}$$

Note that ^{100}Ru to ^{102}Ru are stable.

The discovery of the element promethium (for which the longest-lived isotope has a half-life or only 18 y) in a star (HR 465) in the Andromeda constellation shows that an s-process must be occurring. A possible reaction path is

$$^{146}\mathrm{Nd}\left(\mathrm{n},\gamma\right)^{147}\mathrm{Nd}\left(\beta^{-},11\mathrm{d}\right)^{147}\mathrm{Pm}\left(\beta^{-},2.6\ \mathrm{y}\right) \tag{12.35}$$

The s-process is believed to be extensive in Red Giant stars of mass 3−8 $\mathbf{M}_{\odot}$ and to last for about 10 My, a short period in the total lifetime of a star.

As the star proceeds through the s-process stage we can expect that the fusion reactions decrease while the gravitational contraction of the star continues. To conserve angular momentum the rotational velocity increases, resulting in the ejection of some of the outer mantle into space, thereby exposing the inner, hotter core. Thus in the radio source Cassiopeia A one can see fast moving "knots" of O, S and Ar. The turbulence is likely to bring up heavier elements from the core, as observed in the CP-stars (§12.1.6). For example, the light from the planetary nebula nucleus of FG Sagittae suddenly showed strong barium lines between 1967 and 1972. The turbulence may become so violent that hydrogen from outer layers is mixed in with deeper layers of higher temperature, leading to instantaneous hydrogen burning and a very rapid rise in energy production. This is observed as a sudden light increase from the star (*nova*). After the nova stage (or stages, as several such may occur) the star would continue to cool until it becomes a white dwarf. The density of such a body would be very large, about $10^{8}\ \mathrm{kg\,m^{-3}}$.

12.5.2. Supernova explosions

The s-process cannot explain the formation of the elements heavier than bismuth as the trans-bismuth elements have a number of short-lived isotopes which prevent the formation of thorium and uranium in the amounts observed in nature. The heaviest elements are believed to be formed in supernova explosions.

In a star in which heavier elements are accumulated in its center, the energy production occurs in the layer surrounding the core, see Figure 12.9. For stars of an original

mass >7.5 $\mathbf{M}_\odot$ (or >3.5 $\mathbf{M}_\odot$ at the end of the helium-burning period) the energy loss through photon and, especially, through neutrino emission is very large. This has several consequences:

(i) the emission of energy into space cools the core and the giant begins to contract,

(ii) the atoms in the core can no longer resist the tremendous pressure and their electron shells collapse (forming degenerate matter),

(iii) the contraction and pressure in the core increase to nuclear density (ca. 10^{17} kg/m^3), and the released gravitational energy increases temperature and pressure also in the outer layers.

Under the development of these conditions, the elements in the core disintegrate (especially iron) releasing helium and neutrons; e.g.

$$^{56}\mathrm{Fe} + \gamma \;\rightarrow\; 13\,^{4}\mathrm{He} + 4\mathrm{n} \tag{12.36}$$

The helium immediately fuses and the intense heat developed spreads as a heat shock, which passes to the cooler outer shells of hydrogen and helium, initiating new thermonuclear reactions in the mantle. As a result the whole star explodes as a supernova. While the outer layers expand into space, the core contracts to a *black hole*. Other mechanisms producing supernovae are also known.

A supernovae can be seen by the naked eye as a new star if it occurs within our galaxy. The light intensity slowly decreases and the star may be too weak to see after a year. A supernova discovered by Kepler in 1604 could be seen even in day time and its expanding gas cloud is still easily observed by telescope. The next "nearby" supernova (SN1987A) was seen, on February 24, 1987, in the Large Magellanic Cloud. Before the light arrived to us, a high energy (20−40 MeV) neutrino emission (a total of 19 electron anti-neutrinos was measured in 10 s, compared to a normal 2 per day) was observed in Japanese and American detectors (cf. § 2.5), proving that a large neutrino burst ($\sim 10^{58}$ neutrinos) accompanied the supernova explosion. The neutrinos had an energy corresponding to 5×10^{10} K, and the total gravitational energy released in the process was estimated to 3×10^{46} J.

SN1987A was of a composition like that in Figure 12.9. At the high temperature of the explosion, most of the lighter atoms fused into Fe/Ni, which, thus is not only consumed but also produced "further out". The ^{56}Ni decays according to

$$^{56}\mathrm{Ni}\,(\mathrm{EC}, 6.1\ \mathrm{d})\,^{56}\mathrm{Co}\,(\mathrm{EC}, 77.3\ \mathrm{d})\,^{56}\mathrm{Fe}\,(\text{stable: } 91.7\%) \tag{12.37}$$

The light from the supernova could be related to 0.85 and 1.24 MeV γ-lines from the decay of ^{56}Co; Figure 12.11 shows the decay curve. One could actually calculate that in the first moments after the explosion about 0.1 $\mathbf{M}_\odot$ of ^{56}Ni was formed. Most of the Fe in the Universe is likely produced by this kind of process.

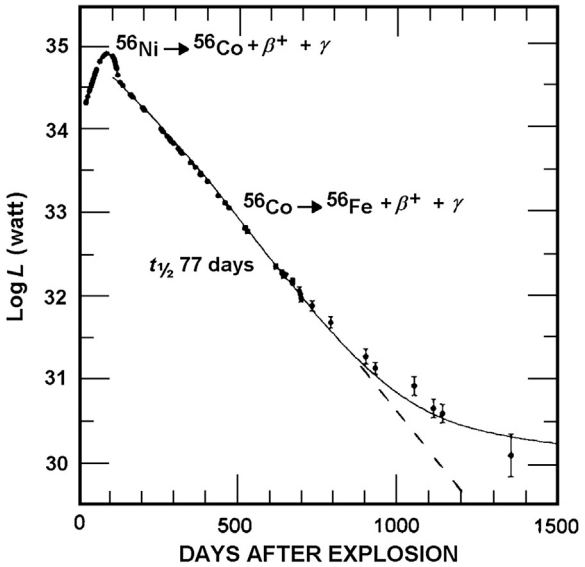


Figure 12.11 Luminosity of SN1987A as function of time. *(From Fransson.)*

The mass of heavy elements spewed out into space enriches it for the formation of later generation stars. The transition elements, which are only formed in such stellar explosions, are a necessity for the existence of life on earth.

12.5.3. The rapid (r-) process

The supernova stage is very short-lived with extremely intense neutron production. It provides a method whereby the barrier of the short-lived isotopes between polonium and francium is overcome and the heaviest elements synthesized. This mode of element formation is known as the *rapid* or *r-process*; see Figure 12.12.

The n-capture in the r-process has been suggested to go up to Z about 100 and $N \leq 184$. In the intense neutron field a considerable amount of (mainly fast) fission of the newly synthesized heavy elements probably also occurs. This partly explains the peaks at $N = 50$ and 82 in Figure 12.10b, which also correspond to maximum yields at $A = 95$ and 140 in thermal fission. Some stars are unique in that they have an unusually high abundance of fission products; spectral lines from heavy actinides, like americium and curium, have also been observed in such stars.

The explosion of thermonuclear hydrogen bombs containing uranium resulted in the formation of elements 99 (Es) and 100 (Fm) (§14.2.9). These elements were synthesized in the extremely short time of the explosion by the intense neutron flux bombarding the uranium (shaded area in Figure 14.5). The explosion of the hydrogen

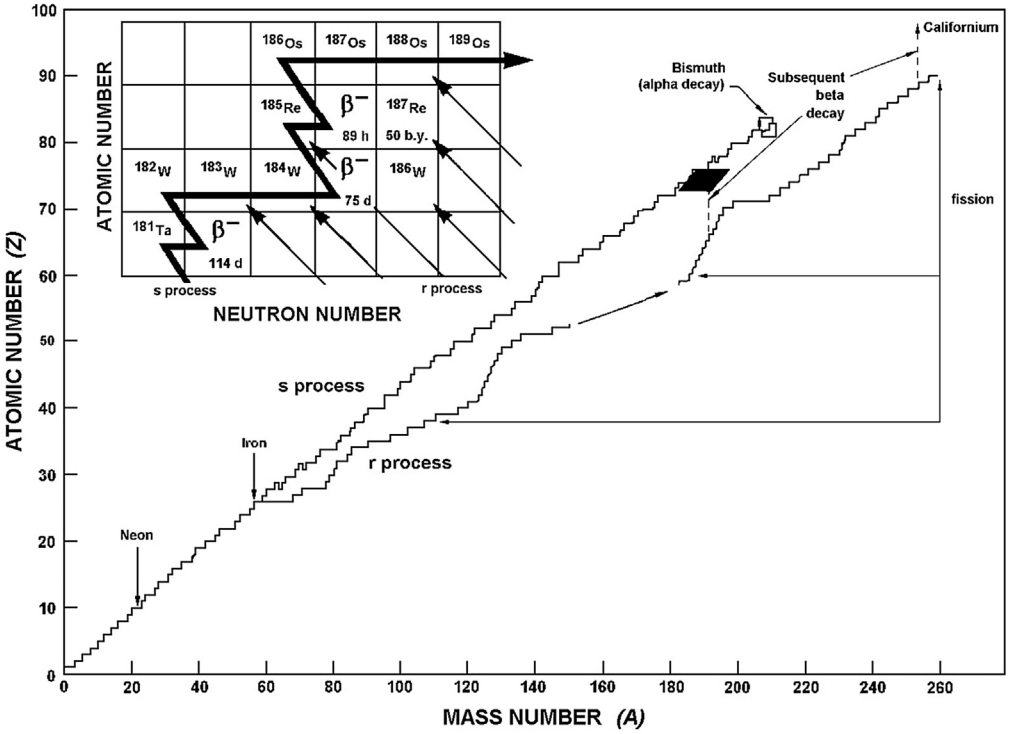


Figure 12.12 Part of isotope chart showing the formation of W, Re and Os isotopes by the s- and the r-processes.

Table 12.4 Comparison of conditions for n-capture processes

Process	Flux (n m^{-2} s^{-1})	×	time	=	exposure (n m^{-2})
s-process	$\sim 10^{18}$		$\sim$1000 y		$\sim 3 \times 10^{29}$
r-process	$>10^{29}$		1 − 100 s		$>10^{29}$
nuclear explosion	$>10^{33}$		<1 μs		$\sim 10^{27}$
nuclear reactor	$\sim 10^{16}$		$\sim$1 y		$\sim 10^{23}$

bomb duplicated in a very small way what is believed to be the process of the formation of the heaviest elements in supernovae. The neutron fluxes and exposures in the s- and r-processes as compared to those in a nuclear explosion and a reactor are given in Table 12.4.

The intensity of the neutron flux as well as the very short time preclude β-decay as a competitor to neutron capture in the r-process. This results in a different isotopic distribution of the elements for the r-process compared to that formed in the s-process. The following reaction sequence illustrates the r-process in which β-decay can occur only

after the explosion has terminated and the intense neutron fluxes decreased (compare with the sequence (12.34)):

$$
\begin{array}{ccccccccc}
^{100}\mathrm{Ru} & (\mathrm{n},\gamma) & ^{101}\mathrm{Ru} & (\mathrm{n},\gamma) & ^{102}\mathrm{Ru} & (\mathrm{n},\gamma) & ^{103}\mathrm{Ru} & (\mathrm{n},\gamma) & ^{104}\mathrm{Ru} \\
 & & & & & & \beta\downarrow 39\ \mathrm{d} & & \beta\downarrow 4.4\ \mathrm{h} \\
 & & & & & & ^{103}\mathrm{Rh} & & ^{105}\mathrm{Rh} \\
 & & & & & & (\mathrm{stable}) & & \beta\downarrow 36\ \mathrm{h} \\
 & & & & & & & & ^{105}\mathrm{Pd} \\
 & & & & & & & & (\mathrm{stable})
\end{array}
\quad
\begin{array}{ccccc}
(\mathrm{n},\gamma) & ^{105}\mathrm{Ru} & (\mathrm{n},\gamma) & ^{106}\mathrm{Ru} & \\
 & & & \beta\downarrow 368\ \mathrm{d} & \\
 & & & ^{106}\mathrm{Rh} & \\
 & & & \beta\downarrow 30\ \mathrm{s} & \\
 & & & ^{106}\mathrm{Pd} & \\
 & & & (\mathrm{stable}) &
\end{array}
\qquad (12.38)
$$

After completion of the r-process, ^{103}Ru, ^{105}Ru, ^{106}Ru undergo β-decay to isotopes of Rh and Pd. In this r-sequence $^{104}\mathrm{Ru} - {}^{106}\mathrm{Ru}$ are formed, but in the s-sequence beginning with ^{100}Ru, the heaviest ruthenium isotope has $A = 103$.

In a supernova explosion a large mass of material is ejected into interstellar space. This contributes to the higher abundance of heavy elements in cosmic rays as compared with the cosmic abundance. In fact, even uranium has been observed in cosmic rays and in our sun. Since our sun is undergoing the simplest type of hydrogen-burning cycle, it is not possible for the heavier elements ($\sim$2%) to have been synthesized by the sun. Consequently, their presence indicates that the sun has been formed as a second (or later) generation star from material that included matter ejected by an earlier supernova, or has accumulated matter from such a star.

The carbon cycle stars are likely to be second generation stars because ^{12}C is needed in the core for the carbon cycle to start. The same star may pass through several novae explosions whereby it loses large amounts of the lighter elements from the outer mantle in each explosion. The chemical composition of a star thus not only indicates its age but also tells us to which generation of stars it belongs.

12.5.4. Neutron stars

A supernova explosion leaves a residue at the center of the expanding gas cloud generated by the explosion. This residue is believed to contract into a neutron star. Assuming a nucleon radius of 0.85 fm, the density becomes about 2×10^{18} (cf. also Figure 4.4 for densities of other nuclear matter). When the central (remaining) mass of the supernova has contracted to a density of $>10^{17}\ \mathrm{kg\,m^{-3}}$ matter will consist of relatively closely packed nuclei and free electrons. It now becomes favorable for the protons to capture electrons

$$\mathrm{p} + \mathrm{e}^- \rightarrow \mathrm{n} + \nu \qquad (12.39)$$

The neutrinos are emitted and only neutrons are left.

The neutron star has a considerable energy reservoir in its rotation, typically one rotation per second. Particles ejected from this rapidly rotating object would be caught in the rotating magnetic field and accelerated to relativistic tangential velocities. They emit high intensity bremsstrahlung, which appear as pulses of radiation with the frequency of the rotation. Such radiation pulses have been observed from cosmic sources known as *pulsars*. The discovery of a pulsar at the center of the expanding Crab nebula, which is the remnants of a supernova explosion observed by the Chinese in the year 1054 has provided strong evidence for the model outlined above. Pulsars have been suggested to be the source of the high energy cosmic ray particles observed in our atmosphere.

12.6. AGE OF THE GALAXY

Estimates of the age of our galaxy comes mainly from theoretical calculations. However, radiochemical investigations can yield information about several important steps. Thus the time since the final solidification of the earth's crust is obtained from the dating of geologic samples (Ch. 13). Nuclear clocks like U → Pb give only the age of minerals; the oldest known minerals on earth are 3.7×10^9 y.

Some glassy materials in meteorites and from the moon are assumed to be original condensed matter which never has undergone melting. The oldest ages seem all to converge at 4.6×10^9 y. Thus we conclude that this was the time when our planetary system began to form (the basis for the estimate of the age of our sun in §12.1.5).

Further information about this event has been obtained by studying tracks which nuclear decay processes leave in certain minerals (§9.1.2). Fission tracks can only persist in minerals that have not been heated because heating above 600°C erases the tracks. The fact that ^{244}Pu fission tracks have been found in iron meteorites and in lunar samples shows that ^{244}Pu existed when the planetary system formed. Because of the short half-life of ^{244}Pu (8×10^7 y) it can be concluded that such mineral samples must have formed within a few hundred million years after the nuclide ^{244}Pu itself was formed. This is probably also the time for planetary formation. The existence of primordial plutonium indicates that an r-process preceeded the formation of the planets.

Apparently, our solar system was showered by debris from a recent but distant supernova explosion which may have disturbed the "peaceful" gas cloud in our part of the universe and initiated the condensation of the solar system. At this time newly formed elements stopped being added to the solar system from the galaxy. Sufficient amounts of ^{129}I and ^{244}Pu remained after the formation of solid materials to produce characteristic isotope anomalies and fission track excesses in meteoritic and lunar materials. The spontaneous fission decay of ^{244}Pu yields ^{129}I, which decays as

$$^{129}\mathrm{I} \xrightarrow[1.6 \times 10^7 \text{ y}]{\beta^-} {}^{129}\mathrm{Xe}\ (\text{stable}) \tag{12.40}$$